The Diagnostics Release Is the Leak:
Controlled Measurement of Revealed Statistics in Multi-Site GWAS

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Abstract

Multi-site genome-wide association studies (GWAS) release not only a final summary statistic but a stream of intermediate diagnostics—allele frequencies, λ_{GC}, the principal-component (PC) spectrum used to correct population stratification, missingness tables. Privacy-preserving pipelines secure the *computation* of these quantities while leaving *what is released* to implementation detail. We measure a question governance proposals assume away: *how much additional membership leakage does each released diagnostic carry?* We make one methodological correction—members and non-members are drawn from the same population, with the attacker's reference panel and the target non-members kept disjoint from the study (and from each other); the common shortcut of scoring the attacker's own reference individuals as non-members inflates measured AUROC by ≈ 0.13. Under this control the two implemented non-degenerate canonical diagnostic scores—the study-fitted PC loading matrix and the allele-frequency summary—are directly comparable: a projection-distance attack on the study-fitted loading matrix (released with the PC eigenvalues) reaches 0.88 AUROC, above the unconditioned allele-frequency summary release (0.84 at both 2 dp and full precision; paired +0.10 for the AF+PC combination, 95% CI $[+0.096, +0.110]$ over 25 seeds). PC outperforms unconditioned AF but is outperformed by conditioned AF (subtracting known calibration members from the released summary), so we state the AF ordering explicitly rather than claiming a universal PC dominance. The other canonical diagnostics we measure (λ_{GC}, the leading-eigenvalue "condition bit" and the missingness vector) enter only as limited controls—a per-release-constant column or an independent random vector—so their near-zero increments limit those abstract constructions rather than ranking realistic λ_{GC}, branch or missingness attacks; we make no ranking against attacks we do not implement. The observed operating point is not a simulator artifact: reproducing the small-$(n \times M)$ regime the paper locates, real 1000G genomes show the same PC>AF ordering within a population (gap +0.05 to +0.14 across twelve real cohorts, and on chr1/chr2) and a scale-matched rerun is consistent with the ordering depending jointly on study size and composition—at matched per-group n the PC channel dominates even across populations, and the AF-first switch appears only as n grows. Among *single* releases, per-site (per-stratum) allele-frequency release is the largest marginal-frequency channel (0.96); two channels outside this canonical set are larger still—an in-sample-fitted haplotype/LD table (0.99) and a membership-correlated missingness curve (whose ρ=1 value reaches 1.00, though a candidate-only baseline matches it, so it is batch association rather than release exposure)—and are reported as separate families below. A *raw* haplotype set is trivially membership-leaking (a member's own haplotypes are present verbatim), so we treat it as out of the governed set. The PC-spectrum leak is attacker-implementable from the release alone (if the study's PC loadings are shared, as federated PCA shares them); its mechanism is the in-sample-ness of the study-fitted spectrum (refitting the loadings on an independent panel collapses the increment to ≈ 0, an independent site's basis separates another site's members only at chance, and a random-basis (eigenvalue-only) projection is at chance while full loadings recover the leak; fixed, data-independent bases are also at chance, including an eigenvalue-weighted fixed basis that reuses the released spectrum (`s53_eigbasis.py`, 0.46–0.53 on simulated and real chr22/chr1/chr2×EUR/EAS/AFR)); a study eigenvalue ratio of 1.000 against an independent sample is consistent with the leak, not evidence against sample-eigenvalue inflation, which both samples would carry. We map a site-size × precision boundary

showing leakage collapses with site size and a coarse rounding cutoff. On real 1000G data the ordering follows the same boundary: within a single population the study-fitted PC channel dominates AF (as it does on the simulated panels), and a scale-matched rerun is consistent with the AF-first switch depending on total study size as well as composition rather than on structure alone. The PC channel also reproduces under a recombination-realistic msprime demography. A distinct haplotype/LD channel reaches 0.99 AUROC and, like the PC channel, is in-sample-only (it transfers to a fresh panel at chance)—the in-sample-ness of fitted released objects is a general property, not peculiar to the PC spectrum. Within a population the PC>AF gap is also density-stable, holds under a stratified permutation null on the same cohorts, and persists to the largest structure-zero cohort the 1000G panel supports (a study/reference split of a single super-population at n_{study} up to 350). We further build and stress-test the logic layer of the governance mechanism itself (a pre-registered reveal manifest, an enforced egress monitor, and a hash-chained cross-run ledger), quantify the model-dependence of the ordering (it is approximately decreasing in measured structure under a size-controlled F_{ST} sweep and crosses at a site-size-dependent measured structure ≈ 0.03–0.15), turn the sanitized-diagnostics paradox into a controlled result with a falsification control, and show a released raw haplotype set leaks membership trivially while a leave-self-out longest-shared-run relatedness statistic does not separate real unrelated members. The implication for pre-registered reveal manifests is direct: the diagnostics that pipelines treat as harmless are exactly the objects that must be governed—and within our tested projection attack, withholding the study-fitted loading matrix removes the leak while a random-basis (eigenvalue-only) release carries none.

1 Introduction

GWAS power comes from sample size, but genotype data cannot be pooled across institutions for legal and ethical reasons. The established engineering answer is to move computation to the data: each site computes summary statistics locally and only the summaries are combined. This rests on an unstated premise—that the summaries *are safe to release*. That premise is false at small site sizes, where allele frequencies and diagnostics can identify individuals.

Crucially, a real multi-site GWAS does not release a single object. It releases a *stream*: per-site allele frequencies, λ_{GC}, the principal-component (PC) spectrum used to correct population stratification, missingness and QC tables, and finally association statistics. Secure-computation work (MPC/HE/TEE federated pipelines) secures how these are *computed*; it says nothing about which of them are *released*. Governance proposals therefore argue that releases should be pre-registered and audited. Such proposals presuppose a fact that has never been measured: *how much leakage does each released diagnostic carry, and does it add anything beyond the final statistic?*

This paper measures that. Our contribution is a controlled leakage-measurement methodology and the finding it produces, not a new cryptographic primitive.

Contributions.

1. **The study-fitted PC loading matrix is a strong intermediate diagnostic leak.** The released study-fitted PC loading matrix supports a projection-distance attack at 0.88 AUROC, above the unconditioned allele-frequency summary release (0.84 at both 2 dp and full precision); combined they reach 0.94, a paired increment of +0.10 (95% CI $[+0.096, +0.110]$ over 25 seeds). A conditioned AF attack—valid post-processing of the released summary using known calibration members, with no curator refit—is stronger still (0.94, above the full-study PC), so we state the AF ordering explicitly: the PC release out-separates the reference-panel AF attack, though not every AF attacker. The remaining canonical diagnostics we measure

enter only as limited controls (per-release-constant columns or an independent random vector), so the comparison is between the PC and AF scores at these operating points. Per-site frequency release is the largest marginal-frequency channel (0.96); an in-sample-fitted haplotype/LD table (0.99) and a membership-correlated missingness curve (≤ 1.00, matched by a candidate-only baseline, so batch association rather than release exposure) are larger still but are separate families (Section 7). The PC>AF ordering also appears on real 1000G genomes in the small-($n \times M$) regime the paper locates: within a single population the study-fitted PC channel dominates AF (gap +0.05 to +0.14 across twelve small real cohorts, where PC is near-saturated at $k/n \approx 1/3$, and a non-degenerate gap +0.10 to +0.19 at n_{study} up to 350 in the fixed-reference pooled arm; Section 9), and a scale-matched rerun is consistent with the ordering depending on total study size and composition (Section 9).

2. **Mechanism identification.** The leak is the in-sample-ness of the study-fitted spectrum—refitting the loadings on an independent panel collapses the increment to ≈ 0 (+0.003), while attacker-realistic centering leaves it intact (+0.10); sample eigenvalues are unchanged (ratio 1.00). An independent site's fitted basis separates another site's members only at chance (0.50), and a random-basis (eigenvalue-only) projection is at chance (0.51) whereas full loadings give 0.88. The attack is implementable from the release alone *if the study's PC loadings are shared*—as federated PCA shares them so each site can project its own genotypes (Section 5).

3. **A site-size $\times$ precision boundary.** Leakage collapses with site size and with a coarse-rounding cutoff; per-site frequency release is by far the worst (Sections 4, 6).

4. **Replication and a second channel.** The PC channel reproduces under a recombination-realistic msprime demography and on two independent real chromosomes (chr1, chr2) at the same sub-AF magnitude with the same in-sample mechanism and loadings-granularity ladder; a distinct haplotype/LD channel reaches 0.99 and shares the in-sample-only property (Section 7).

5. **A built governance layer.** A declarative reveal manifest, an egress monitor, and a hash-chained ledger are implemented and stress-tested at the release-object level; the ordering's model-dependence is quantified (approximately decreasing in measured structure under a size-controlled F_{ST} sweep, crossing at a site-size-dependent measured ≈ 0.03–0.15), the sanitized-diagnostics paradox is upgraded to a controlled result with a falsification control, and the tested IBS/LSR channel fails to separate unrelated real members (Section 8).

6. **A governed-object implication** and the honest scope of the governance mechanism (Section 10).

2 Related Work

Secure GWAS computation—MPC, homomorphic encryption, and TEE federated pipelines—secures the *computation* layer [7, 1, 12, 3]. secure, collaborative GWAS systems now realize full PCA-based and LMM-based federated pipelines with cryptographic guarantees [16], releasing the study-fitted PC *loadings* so each site can project its own samples. Our measurement takes that released object as given and asks what it leaks. Recent work also shows released PCs support full *data reconstruction*, a strictly stronger attack than membership [17], and that GWAS summary statistics combined with high-dimensional phenotype data enable genotype recovery and individual re-identification under a

linear-programming formulation [18]—so the release object we measure sits inside an active leakage literature we do not claim to originate. Release-safety work assesses whether a *single* final release is safe [8, 9]; interdependent-release work checks combined releases of selected SNPs [9]. The measurement substrate is the genomic membership-inference literature [4, 10, 11, 2], and recent work warns that reported MI gains can reflect shared cohort traits rather than membership [19]—the confound our disjoint-negative control addresses. Outside genomics, a released PCA is already known to leak membership directly: Zari et al. [15] attack the principal components through a reconstruction-error score and report stronger leakage for smaller training sets and more components. Our PC result is not a new attack primitive; we reuse that released-object leakage in a controlled GWAS setting, and our contribution is the *comparison* of the intermediate simulated/released objects at matched operating points (the PC-vs-AF ordering and its site-size/structure boundary), the requirement that the release name the *loading matrix* as the leaky object, and the governance layer over the release set. Genomic privacy *metrics* are known to be non-monotone under adversary strength and cannot anchor a security claim [14], which is why we report model-specific empirical estimates rather than a budget. Workflow provenance and transparency logs [6, 13, 5] record what happened but cannot pre-authorize an output. Our contribution sits between these: a controlled measurement of the *intermediate* releases that all of them leave unmeasured.

3 Threat Model

We consider three adversaries. **A1: honest-but-curious site**—follows the protocol, tries to infer others' inputs from protocol records; handled by the MPC layer's simulation security and *not* a claim of this paper. **A2: constrained workflow author**—may under-declare or over-precision releases, use the wrong recipient, or loosen a release condition, but cannot execute arbitrary code or bypass the mediated runtime. **A3: adaptive analyst**—legitimately submits several analyses and uses earlier releases to build later ones, accumulating leakage; we measure this but make *no* formal composition claim. We measure A2/A3-visible releases under a membership-inference adversary holding a disjoint reference panel. We make *no* data-independent privacy claim; every number is a model-specific estimate. We separate the syntactic layer (what is released) from the empirical layer (what it leaks).

4 Controlled MI Protocol

Data. We simulate a three-population structured panel (Balding–Nichols, $F_{ST} = 0.08$) and, as an independent, recombination-realistic cross-check, an msprime three-population demography over 2 Mb. Each population is split three ways, disjointly: study members, attacker reference panel, and target non-members. QC keeps variants with minor-allele frequency in $[0.05, 0.95]$ (≈ 3800 of 4000 simulated sites per panel).

Attack. An allele-frequency likelihood-ratio membership score ($\sum_j \log P(g_{ij} \mid \hat{f}_j^{\text{study}}) - \log P(g_{ij} \mid \hat{f}_j^{\text{ref}})$). As a structure-only control we project both target panels onto an independent reference panel's PC basis and score proximity to its centroid; because both target panels are out-of-sample with respect to that basis and ancestry-matched to each other, this control isolates the pure population-structure channel.

Operating point. Unless stated otherwise, each experiment uses three ancestries of 300 individuals each per panel (900 study members, 900 attacker reference, 900 independent non-members

Table 1: Release-precision frontier (disjoint non-members, 5 seeds). The ancestry/structure-only control—both target panels projected onto an independent reference panel's PC basis—is at chance (0.502) at every tier; per-ancestor AUROC is uniform (F5: 0.860/0.849/0.851). The last column is the *mean of the per-seed bootstrap bounds for TPR@1%*, not a confidence interval for the 5-seed mean.

Tier (what is released)	AUROC	TPR@1%	95% CI
F2 top-200 SNPs	0.590	0.023	[0.009, 0.044]
F4 all SNPs, AF 1 dp	0.622	0.031	[0.014, 0.054]
F3 all SNPs, AF 2 dp	0.840	0.183	[0.119, 0.255]
F5 all SNPs, AF full	0.845	0.193	[0.132, 0.254]
F5 per-site AF	0.964	0.588	[0.490, 0.696]

from the same three ancestral populations) and 4000 simulated sites ($\approx$ 3814 passing QC; site-to-member ratio $\approx$ 4.2). Real-data numbers use the 1000 Genomes panels described in Section 6. Seeds are fixed per experiment (5 for the frontier and mechanism, 3 for the intermediate-reveal table, 2 for the boundary and dimension sweeps; the paired-significance study uses 25 seeds, the real-data orientation and mechanism studies 20 splits, the crossover map 12 seeds per cell, the multi-chromosome replication 20 splits per chromosome, the site-size sweep 8 seeds per cell, the msprime/haplotype studies 5 seeds, the fixed-basis probe / second-haplotype / real size-ladder studies 5 splits or seeds per cell, the F_{ST} sweep 15 seeds per cell, and the admixture study 10 seeds per setting); uncertainties are reported as bootstrap CIs where the effect is small.

Methodological note: the self-reference bias. The common shortcut scores the attacker's own reference individuals as non-members. Because those individuals are pulled toward $\hat{f}^{\text{ref}}$, the test becomes a symmetric two-panel distinguishability test and overstates the effect. Quantified directly on the same panels (5 seeds, full-precision AF release), the biased protocol yields AUROC 0.976 versus 0.845 for the disjoint protocol—an inflation of 0.132 ($\approx$ 0.13 AUROC). We therefore use a third, disjoint non-member panel and report corrected numbers throughout; this control-design caution is one of the paper's methodological takeaways.

5 The Diagnostics Release Dominates

We write the release tiers of Table 1 as $\mathbf{F}n$ (final summary release at increasing precision/visibility) and the intermediate diagnostics as $\mathbf{M}n$ (M0 none; M1 λ_{GC}; M2 PC spectrum; M3 a random missingness vector; M4 the leading eigenvalue as a stand-in for a hidden condition bit). "Intermediate" means released *before* the association statistic, as a workflow diagnostic; the per-site frequency release (F5 per-site) is itself an intermediate object but is reported separately as the largest marginal-frequency single channel. We add the PC-spectrum diagnostic to the attack as per-target features (projection distance to the study centroid, and |PC1| alignment), evaluated by a cross-validated logistic meta-attack with disjoint negatives. Among these *implemented* scores the PC channel is the only one that moves the attack; M1 (λ_{GC}) and M4 (the leading eigenvalue) are per-panel constants—identical for every individual in a release—so they cannot enter a per-individual classifier, and M3 is an independent random vector, so its feature is a random-projection control. The negligible increments of M1/M3/M4 therefore show that *these constructed controls* add nothing to this attacker, not that a realistic λ_{GC}-, branch-, or missingness-based attack is weak.

Table 2: Marginal leakage of intermediate reveals (disjoint negatives, 3 seeds). M1/M4 are per-panel-constant columns and M3 is a random-vector projection—limited controls, not realistic attacks of those channels. The PC-spectrum increment is confirmed at 5 seeds (Section 5).

Feature set	AUROC	Δ vs final-only
final release only	0.844	—
$+\lambda_{GC}$ (const)	0.844	0.000
+ eig1 (const)	0.844	0.000
+ missingness (rand)	0.844	0.000
+ PC loadings	**0.942**	**+0.098**

Mechanism. Projection distance alone gives 0.881 AUROC—above both AF summary releases (0.840 at 2 dp, 0.845 at full precision); |PC1| alone is at chance (0.501). Refitting the loadings on the attacker's reference panel (both target panels then out-of-sample) collapses the increment to ≈ 0 (+0.003). Centering all targets by the public study mean leaves the increment intact (+0.09 to +0.12, mean +0.10). The top-10 eigenvalues of the study and an independent sample are equal (ratio 1.000), which shows the release does not report inflated study eigenvalues relative to an independent sample; it does not by itself exclude sample-eigenvalue inflation (both samples could carry it), so we attribute the leak to the study-fitted *directions*, a claim the transfer test above supports directly. Note the sign: members sit systematically *farther* from the study centroid than non-members, because the study-fitted PCs are tuned to the members' variance— so the study-fitted PC-distance attack scores ≈ 0.12 with the naive orientation and 0.88 with the flipped orientation (the meta-attack learns the sign, as an adaptive adversary would); we report it as 0.88, which is the orientation an adversary would use. A within-ancestry label-permutation control returns chance (0.486).

Paired significance and an auxiliary-information control. Re-running the comparison on 25 seeds with the orientation calibrated on a *held-out* half of the labels (so the reported test AUROC uses no test-side label information), the calibrated PC channel beats the AF summary release on every seed (0.879 ± 0.009 vs 0.832 ± 0.021; paired mean gap +0.046, 95% CI $[+0.037, +0.055]$; paired t / Wilcoxon / sign tests all $p < 10^{-4}$), and the logistic combination beats AF by +0.103 (95% CI $[+0.096, +0.110]$). The train-calibrated orientation is the members-farther direction on 25/25 seeds, agreeing with the mechanism paragraph above.

The calibrated PC channel and the calibrated logistic combination both use the known membership of the train half, so we also give the AF channel the same known-*member* side information: subtract the known calibration members' genotypes from *the released* study AF and score against the residual frequency for both classes with disjoint negatives (`s10_paired.py`, `af_conditioned`). This is valid attacker post-processing of the original release, not test-label leakage. It reaches 0.941 ± 0.012, *above* the full-study calibrated PC (0.879 ± 0.009) on 25/25 seeds, and above the logistic AF+PC combination (0.935). So within this S10 comparison the conditioned AF attack has the highest mean AUROC among the attacks on the pooled AF release and the original PCA release (it beats the combination on 19/25 seeds—a higher mean, not an every-seed advantage), and the headline PC>AF ordering is against the *unconditioned* AF baseline (+0.046, 95% CI $[+0.037, +0.055]$). We do not extend the conditioned-AF result to the real-data experiments, where it was not evaluated.

A natural follow-up refits the PCA on the same remaining panel; that is a *curator* action (it changes the released object, not an attacker post-processing) and is a smaller-cohort release

comparison, reaching 0.964 ± 0.006 (`pc_conditioned`; the orientation is reused from the training-half calibration, positive on every seed). It is not an attack on the original release: the residual mean is reconstructable from the released statistics, but the refitted loading matrix is not, so this comparison does not spend the same attacker information as the AF residual. The member-only leave-own-out residual (subtracting the scored member's own genotype before scoring it) is not a deployable attack—choosing that branch uses the membership being predicted—and collapses to 0.429; we report it only as a mechanism reading.

The leak is in-sample-ness, not a transferable structure channel. Splitting the study into two disjoint sites A and B and scoring each site's members against the shared non-members, a site's release-fitted basis separates its *own* members at 0.88 but the *other* site's basis separates them only at chance (0.50); a swapped null (scoring site B's members with site A's basis) is likewise at chance (0.51). The channel is the released basis being fit on the very individuals it is used to test; it does not transfer to a fresh site. The same test on *real* 1000G chr22 (20 four-way splits of two disjoint sites plus a shared non-member set) reproduces it: a site's own basis separates its members at 0.685 ± 0.015, but the other site's basis gives 0.516 ± 0.013 and a swapped null 0.516 ± 0.010—so the in-sample mechanism holds on real data as well, at the smaller real-data magnitude.

The leak needs the loadings, not the eigenvalues. Releasing only the top-10 eigenvalues (with an attacker-substituted random orthonormal basis) leaves this projection channel at chance (0.51); releasing top-k loadings recovers it monotonically (k=1: 0.53, k=3: 0.65, k=5: 0.74), and the full top-10 loadings give 0.88. Withholding the loading matrix removes this projection attack; the random-basis result is a specific (eigenvalue-only projection) ablation, not a proof that no eigenvalue-dependent attack exists, and fixed, data-independent bases—including one weighted by the released eigenvalue spectrum—are also at chance (0.46–0.53 on simulated and real chr22/chr1/chr2×EUR/EAS/AFR), so a purely eigenvalue-informed fixed basis does not recover the leak.

The structure control is not a no-op. The reference-fitted projection control returns 0.502 on ancestry-matched members/non-members—its expected null—but rises to 0.63 when the same statistic is asked to separate two genuinely different ancestry groups, confirming it responds to a real structure signal rather than being inactive.

Dimension dependence and the PC−AF crossover. We map the paired increment (PC−AF, PC orientation train-calibrated) over a per-population-group × marker-count grid (Fig. 1(c); the population group has 400/800/1600 simulated individuals, from which the study panel is a 0.3 fraction per stratum, i.e. 360/720/1440 study individuals over three strata). The increment is positive throughout, peaks near the sparse-marker/small-panel corner ($+0.16 \pm 0.04$ at 360 study individuals, 500 markers) and shrinks toward denser marker sets, crossing to ≈ 0 only at the far corner (1440 study individuals, 8000 markers: -0.01 ± 0.01), where the AF attack is already near saturation at 0.86. The PC channel thus overtakes AF in the small-panel regime and loses to it only when the AF attack is already near saturation—consistent with the earlier nine-point sweep, now with the crossover located explicitly.

Robustness across functional forms. The frontier is not specific to the likelihood-ratio functional: on the same panels, a correlation-form attack (0.833) and a cross-validated learned-deviation attack (0.832) track the AF-LRT (0.845) at full precision, and at 1 dp the correlation/deviation

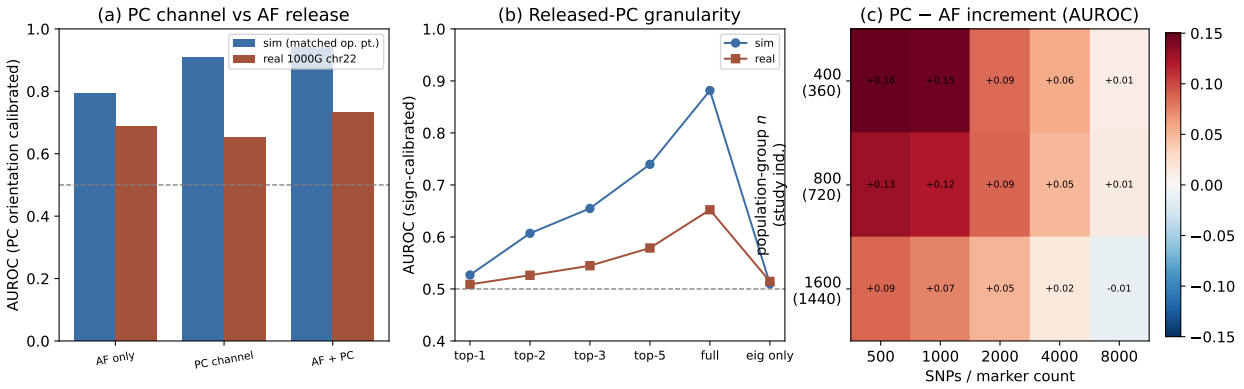


Figure 1: The PC channel. (a) Orientation: on the matched simulation the calibrated PC channel dominates AF; on real 1000G chr22 it is smaller than AF and its sign-calibrated value sits below the AF release. (b) Released granularity: the sign-calibrated AUROC rises with the number of released loadings and is at chance for an eigenvalue-only (random-basis) release, on both simulation and real data. (c) Paired PC−AF increment over the panel-size × marker grid; positive in the small-panel regime, crossing to ≈ 0 only when AF saturates.

Table 3: TPR@FPR≤1% by individuals-per-site and AF precision (disjoint negatives).

n/site	AF 1 dp	AF 2 dp	AF full
50	0.270	0.993	0.993
100	0.183	0.745	0.723
200	0.062	0.307	0.301
400	0.040	0.133	0.136
800	0.015	0.038	0.042

forms retain more signal (0.660/0.659) than the LRT (0.622)—the LRT is the more rounding-fragile functional. The PC-diagnostic channel (0.882) remains above all three at every tier.

6 Site-Size and Precision Boundary

Leakage collapses with site size and there is a coarse/fine cutoff: rounding to 1 dp reduces it sharply while full and 2 dp precision behave nearly identically, so the operative lever is the cutoff, not unbounded precision. An independent msprime demography (weakly structured, 2 Mb, recombination-realistic) reproduces the level (AUROC 0.642 at full precision, 0.649 at 1 dp) but not the rounding benefit: at that operating point the quantization effect is within noise. We report this honestly—it is consistent with the saturation reading, in which the benefit comes from crossing the coarse cutoff rather than from finer rounding at the structured-population operating point.

Real data. The same protocol on *real* released allele frequencies (1000 Genomes Phase-3 chr22, three super-population pseudo-sites EUR/EAS/AFR, each split three ways disjointly, MAF $[0.05, 0.95]$, distance-thinned at a 15 kb window; averaged over 5 random three-way splits) reproduces the ordering with smaller magnitudes: AUROC 0.684 (full AF), 0.681 (2 dp), 0.564 (1 dp), 0.640 (top-200), and 0.808 for per-super-population release (±0.009–0.014 across splits); the reference-basis structure control is at chance (0.504). The qualitative frontier—per-population > full/2 dp > 1 dp/top-

k—holds on real data. We note honestly that the smaller real-data magnitudes do *not* have a simple explanation: the real chr22 panel is, if anything, *more* structured than the simulated panels (between-super-population F_{ST}-like statistic ≈ 0.10 on the full chromosome vs ≈ 0.05 simulated; chr1 and chr2 are higher still, ≈ 0.14 and 0.11; the top-10 PCs capture 0.26 of total genetic variance vs 0.14 simulated; both computed in `s15_controls.py` and `s20_extra.py`, the latter on a 2000-marker subset), so "the real panel is less structured" is the wrong explanation. The plausible cause is that the real top PCs are dominated by super-population structure that members and non-members share equally, leaving a smaller *in-sample-ness* share for the attack to exploit; we flag this as a hypothesis rather than a measured result.

Real-data PC channel. The released study-fitted PC channel is *present but sub-dominant* on real data, unlike in the simulation. Running the S9 orientation study on real 1000G chr22 at a matched operating point (450/panel, 1363 SNPs) over 20 splits: the calibrated PC channel reaches 0.652 ± 0.014 and the AF-only release 0.688 ± 0.026—so the PC channel sits *below* AF on real data, where in the matched simulation the calibrated PC channel is 0.910 and AF 0.794. The direction of the PC channel is the same on real data as in simulation (members sit *farther* from the centroid; the calibrated-vs-naive orientation is the same on 20/20 splits), so the discrepancy is magnitude, not direction. The granularity structure also replicates: on real data the calibrated channel rises with the number of released loadings (k=1: 0.509, k=3: 0.545, k=5: 0.579, full 0.652) while an eigenvalue-only (random-basis) release is at chance (0.515)—the tested projection leak needs the loading matrix on real data too, it is simply weaker. Added on top of the AF release, the real-data PC channel gives a small but consistent increment ($+0.043$, 95% CI $[+0.033, +0.053]$, positive on 20/20 splits, paired t-test $p = 1.3 \times 10^{-7}$; the two channels are partly redundant, so the increment is smaller than either channel alone). Combined with the real-data mechanism test above, the picture is consistent: the same in-sample-ness mechanism, at smaller magnitude at the mixed real operating point (a magnitude difference governed by study size and structure, Section 9). We therefore scope the claim: *relative to the unconditioned AF-LRT baseline, the diagnostics release (the study-fitted PC loading matrix) is the stronger of the two implemented non-degenerate canonical diagnostic scores in the small-($n \times M$) regime—which includes single-population real cohorts, where the PC diagnostic dominates the AF release—while once study size and structure grow the AF release leads*, and we report the operating-point dependence rather than tuning it away. (The real-data top-200 tier is selected by $|\text{AF} - \text{refAF}|$ rather than by association p-value, because the real release has no phenotype; this is a proxy for a released interesting set.)

7 Replication and a Second Channel

We stress the two scoping caveats—real data is chr22-only, and the PC channel was only measured on the parametric simulator—by replicating on two independent chromosomes and on a recombination-realistic simulator, and by adding a genuinely distinct attack family.

Two independent real chromosomes. Running the orientation, mechanism, and granularity studies on chr1 (5257 markers) and chr2 (4664 markers), each with 20 disjoint three-way splits (Fig. 2(a)): chr1 gives AF 0.768 ± 0.045, calibrated PC 0.675 ± 0.016, AF+PC 0.805; chr2 gives AF 0.752 ± 0.046, PC 0.689 ± 0.015, AF+PC 0.799. The chr22 ordering replicates: on *both* new chromosomes the PC channel is *below* the AF release but adds a consistent increment. The mechanism also replicates—a site's own basis separates its members (chr1 0.73, chr2 0.74) while the other site's basis is at chance (0.51) and a swapped null is at chance (0.51)—and so does the granularity lad-

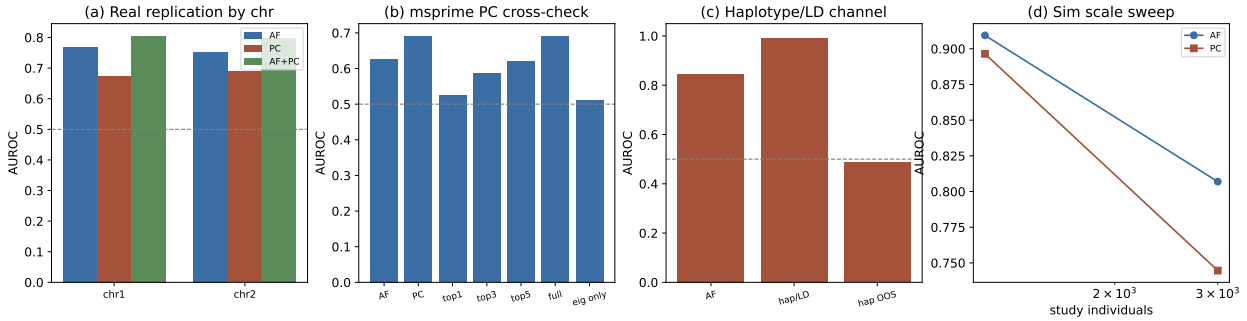


Figure 2: Replication and a second channel. (a) Real replication on chr1 and chr2: the PC channel is below AF but adds a consistent increment, replicating the chr22 ordering. (b) msprime cross-check: PC above AF with the same granularity ladder. (c) A haplotype/LD channel reaches 0.99 but transfers to a fresh panel at chance—in-sample-only, like the PC channel. (d) Site-size sweep (M=10,000): the +PC increment is stable (+0.06) but PC no longer dominates AF at $n \geq 1200$.

der (full 0.675/0.690, monotone in k, eigenvalue-only 0.516/0.514). That the same in-sample-ness mechanism and loadings-granularity ordering hold on three independent chromosomes strengthens the real-data reading beyond the single chr22 result.

Recombination-realistic simulator. On an msprime three-population demography (real recombination, 5 Mb, realistic site- frequency spectrum), the PC channel is again above AF (calibrated PC 0.691 ± 0.013 vs AF 0.627 ± 0.015; Fig. 2(b)), with the same granularity ladder (top-1 0.524, top-5 0.621, full 0.691, eigenvalue-only 0.512). The held-out-site mechanism replicates under an ancestry-matched within-population split (in-sample 0.729 ± 0.015, out-of-sample 0.510 ± 0.010, swapped null 0.518 ± 0.012). The PC-dominance and the mechanism are therefore not artifacts of the parametric simulator's injected LD blocks.

A haplotype/LD channel. Our AF-LRT, correlation, and learned- deviation attacks all consume per-SNP marginals. A distinct family uses the *joint* distribution of adjacent markers (the within-window haplotype structure): score a target by $\sum_j [\log P_{\text{study}}(g_j, g_{j+1}) - \log P_{\text{ref}}(g_j, g_{j+1})]$ over adjacent pairs, at the same operating point (900/900/900, 4000 markers, 5 seeds). This reaches 0.990 AUROC—above every marginal channel—and, crucially, it is *in-sample-only*: fit on disjoint panels that saw neither the study nor the targets, it transfers at chance (0.487; Fig. 2(c)). The *released joint LD table* is a hypothetical release object (a workflow could publish pairwise r^2/haplotype frequencies as a QC diagnostic), so we report it as a family bound rather than a standard release. The in-sample-ness of a fitted released object is thus a general property of the release, not peculiar to the PC spectrum. A leave-self-out longest-shared-run relatedness diagnostic is reported in Section 9, where it is at chance on three chromosomes at physical length.

The site-size regime. At realistic site sizes with 10,000 markers and 8 seeds (Fig. 2(d)), the PC channel's dominance over AF does not survive: at $n = 1200$ it is roughly tied (AF 0.909, PC 0.896) and at $n = 3000$ AF leads (AF 0.807, PC 0.745). The +PC increment is essentially unchanged (+0.055–+0.056), so the *combination* stays above AF (0.966/0.862) and the PC channel still contributes—but the ordering statement is genuinely an operating-point property. This is consistent with the crossover map: PC-dominance is a small-site phenomenon, and at the site sizes real biobanks report the AF release already saturates.

8 Building and Stress-Testing the Governance Layer

The paper argues for pre-registered reveal manifests, enforced egress, and cross-run auditing; here we build *both* the release-object layer and the sole-egress binding of that mechanism and stress-test them. The prototype (`code/governance.py`) provides (i) a declarative, content-hashed, versioned *manifest* of the quantities a run may release, each with a semantic type, precision, recipient, and condition; (ii) an *egress monitor* that wraps the study's release objects and can emit only declared quantities, coerced to the declared precision (decimal rounding, top-k truncation, binning, threshold-bool); and (iii) a tamper-evident *hash-chained ledger* storing a value hash per release (never raw values) plus the cross-run relations refinement / condition-loosening / duplicate / recipient-expansion.

A built sole-egress binding. The monitor above is a plaintext library call: a study author who simply does not call `release` writes a withheld quantity straight out. We first demonstrate the bypass as a positive control (`code/s39_enforcement.py`): the same exfiltration line (`open(...,'wb').write(loadings)`) succeeds against the library prototype and the withheld loadings land on disk (3/3 seeds). We then close it with a minimal process-isolation runtime (`code/enforce.py`): the study computation runs as a child process whose singleton sanctioned egress route is one pre-opened pipe descriptor wired to the parent's `EgressMonitor`; its stdout and stderr are discarded (`/dev/null`), and a startup PEP-578 audit hook denies write-opens, socket creation and datagram/stream sends, subprocesses, and `ctypes` loading. Under the isolated runtime the same natural-egress write is blocked (3/3 seeds, the leak file is never created), a sanctioned push of an undeclared object is blocked by the parent monitor (the second enforcement point), a declared release still flows and its hash chain verifies, and the transported governed release lands at the §8 governed tier (AF 0.624, PC 0.509). A bounded regression check closes the ordinary-Python datagram route: the same runtime that blocks `socket.connect` also denies `socket.__new__` / `sendto`, and a child that tries an ordinary UDP `sendto` to a fixed `127.0.0.1:9` destination is stopped at socket creation ("UDP sendto returned (not blocked)" never observed, 3/3 seeds, `s39_enforcement.py`). The monitor's release-object API handles the per-group (dict-valued) AF object directly: permitted full-precision and rounded (2 dp) per-group AF releases both flow through `EgressMonitor.release` with the ledger chained (3/3 seeds), so the strongest single bundle channel is supported at the release-object API level. The *isolated* transport is tested on a fixed declared set (AF, eigenvalues, λ_{GC}) and on blocking an undeclared object; it does not yet emit per-group AF through the monitored release path (per-group AF is serialized and restored in the parent, but is omitted from the isolated runtime's monitored release loop). The threat model is the careless study-code author (ordinary Python, not root, no native code), matching the paper throughout; a local user with process control and the parent runtime stay in the trusted computing base. The stated "no native code" boundary is itself tested: an injected `ctypes.CDLL(None)` call from the child is denied on `ctypes.dlopen` / `ctypes.dlsym` (3/3 seeds), so the native loading route does not escape the hook. This turns the one previously argued mechanism into a built and tested one.

Enforcement. Under a governed manifest (AF at 1 dp, eigenvalues released, loadings/af-by-group/missingness/p-values withheld) the monitor's outcome matrix on the standard attack injections is: undeclared loadings, undeclared missingness, undeclared condition bit, recipient expansion, and condition loosening, all *blocked*; an attempt to emit full-precision AF into a 1 dp slot is *coarsened*; the hash chain verifies (`s23_governed.py`). The consequence is the predicted one: the leakage

attacks on the governed release drop from AF 0.845 / PC-dist 0.882 to AF 0.622 / PC-dist 0.510 (Fig. 3(b))—i.e. the monitor's output equals the governed tier.

Absolute per-individual exposure. Every number above is a rank statistic (AUROC); a privacy reviewer asks what it means for one person. We re-express the operating points in absolute per-individual terms (`s38_privacy_metric.py`, 5 seeds): the membership advantage $A^* = \max_t |\mathrm{TPR} - \mathrm{FPR}|$ (the two-sided KS distance between the member and non-member score distributions), the per-individual posterior ρ_i from a score-density likelihood ratio, and a resolution-floored empirical f-DP trade-off summary $\varepsilon_{\mathrm{lb}}(\delta) = \max_\alpha \max\{\log((\mathrm{TPR}(\alpha)-\delta)/\alpha),\ \log((1-\delta-\alpha)/(1-\mathrm{TPR}(\alpha)))\}$ over the two non-trivial branches of the trade-off function $g_{\varepsilon,\delta}$. It is an in-sample, resolution-floored summary of the observed trade-off, evaluated over rates at or above their sample resolutions ($1/n_{\mathrm{neg}}$=$1/n_{\mathrm{pos}}$=1/900) and computed on the same individuals the score was fit on, so it is a descriptive, resolution-limited summary of the observed trade-off rather than a release-level guarantee; a false-negative rate floor ($1/n_{\mathrm{pos}}$ in the default reading, or $3/n_{\mathrm{pos}}$—the rule-of-three upper bound on a zero count—in a conservative reading) keeps the strict trade-off finite where a single zero observed false-negative count would otherwise make it infeasible, so the reported values are floored under a stated convention (the `valid` flag checks consistency with that convention, not statistical validity); we do not model aggregate k-anonymity, which is not well-defined for these releases. The selected comparisons retain their ordering: the released loadings give A^*=0.61 and $\varepsilon_{\mathrm{lb}}\approx$5.0 at δ=0.01 (the quoted ε values use the default $1/n_{\mathrm{pos}}$ floor), versus 0.55/4.1 for full-precision AF and 0.19/1.1 at 1 dp; the reference-fitted-basis and eigenvalue-only controls sit at $A^*\approx$0.04, $\varepsilon_{\mathrm{lb}}$=0.19–0.24 (the ε summary is not monotone in AUROC across all channels—full-precision AF has the higher AUROC over AF@2dp but the lower ε—which is why we read it per-channel rather than as a global ordering). A^* is the KS distance, a distinct quantity from AUROC rather than a rescaling of it. The metric is not a simulator artefact: re-expressed on *real* 1000G chr22 and on the semi-synthetic cohort (8 splits; 3 cohort seeds, `s44_individual_real.py`), the ordering is preserved (real chr22 PC loadings A^*=0.27 below full-precision AF 0.35 and sitewise AF 0.58, matching the real AUROC ordering; on the cohort AF 0.13 leads PC 0.04) and both controls sit at $A^*\leq$0.07. The ε summary is descriptive and floored under a stated convention (default $1/n_{\mathrm{pos}}$; conservative rule-of-three $3/n_{\mathrm{pos}}$) on every channel, so we read it per-channel rather than as a statistical bound.

Composition. Individually manifest-permitted releases compose: a run sequence (λ_{GC}, eigenvalues, AF@1dp, AF@2dp, AF full, then the loadings) has cumulative AUROC 0.941 vs 0.882 for the largest single release, and the union exceeds the max constituent in 5/5 seeds (Fig. 3(c); the constant λ_{GC}-only first run sits at the chance baseline 0.50, as does its single-release point). The ledger records a refinement relation at the AF→loadings step; this is *producer-asserted* metadata (the prototype records and audits relations, it does not content-infer them), so an anti-composition policy turns the declared refinement into a pre-registered violation rather than the monitor silently inferring it.

A multi-object bundle leaks more than its parts. The composition result above is a *declared sequence*; a realistic run releases a bundle at once. Measuring the full aggregate of a constructed bundle (pooled AF@2dp, per-site AF@2dp, top-10 eigenvalues, λ_{GC}@2dp, the per-SNP call-rate table, top-50 p-values) with a joint out-of-fold logistic attacker (`s40_bundle.py`, 5 seeds; the classifier is trained on four of five stratified folds with known membership and scored on the held-out fold, a separate split from the half-study compositional split that defines the call-rate table's contributor set: the call-rate table the study releases is the object the attacker consumes, coarsened

at the manifest's declared 2 dp), the bundle reaches 0.958 versus 0.947 for the strongest single object (per-site AF), an excess of +0.011 in 5/5 seeds. Leave-one-out attribution shows per-site AF is the largest single contributor (dropping it costs 0.084), and the excess comes from the *orthogonal* per-individual channels it combines (per-site AF plus the call-rate table), not from feature count: stacking a random per-individual column onto pooled AF leaves the AUROC unchanged (0.839); the per-panel-constant objects (eigenvalues, λ_{GC}, p-values) carry no per-individual variation and contribute no feature to the evaluated per-individual attacker (they could still carry candidate-dependent signal in an attacker we do not implement, so the negative is scoped to the tested attacker). The bundle is a constructed composition example: its objects do not all share one contributor set. The genotype-derived objects (pooled AF, per-site AF, eigenvalues, p-values) are fit on the full study; the call-rate table is averaged over a train half of the study plus all reference individuals (the contributor split is exposed through the M3 contributor/non-contributor reading below). The measured excess is therefore an attribution of which object adds signal, not a claim that a single workflow emits exactly this contributor composition. The released call-rate table adds a small conditional increment beyond the candidate's own call-count information: AF channels plus the candidate-only call count reach 0.952, and adding the released-table M3 score raises it to 0.959 (+0.008, positive 5/5 seeds; the second manifest gives 0.965→0.971, +0.006, 5/5), so the released table carries a small gain over the AF channels plus the candidate's total call count (a specific comparator, not all candidate-side information). The M3 call-rate feature scores every member (the released study table is observable to all of them); restricted to the members who did *not* contribute to the table its AUROC is 0.511, so the M3 channel's 0.705 is largely the members who are themselves averaged into the released table—we report both readings. The excess is not specific to that manifest or to the coarsened comparator: under a second manifest (per-site AF at *full* precision, eigenvalues, λ_{GC}@2dp, call-rate table; pooled AF and p-values omitted, `s46_bundle_manifest2.py`) the bundle reaches 0.970 against an *uncoarsened* strongest single object at 0.964, an excess of +0.006 (positive 5/5 seeds; +0.023 against the coarsened single object of the first manifest). So the per-channel analysis under-estimates the bundle, and an anti-composition clause must cover orthogonal channels, not only the AF-versus-loadings refinement.

Model-dependence of the ordering. The PC-vs-AF ordering is not free-floating: sweeping the simulated F_{ST} from 0.02 to 0.24 (three site sizes, 15 seeds), the PC−AF gap decreases at every site size and crosses zero at a *measured* structure statistic that depends on site size (the size-resolved crossing brackets are $[0.027, 0.052]$ at n=1600, $[0.052, 0.077]$ at n=800, $[0.103, 0.151]$ at n=300); a within-population real cohort (Fst ≈ 0) sits *before* the crossing, and the real mixed three-population cohort (measured Fst ≈ 0.09 at n_{study}=450) sits near parity, i.e. *at* the crossing for its size rather than clearly beyond it (Section 9). The read is best done in *measured* terms: at nominal F_{ST}=0.08 the realized between-group statistic is ≈0.05, i.e. measured ≈ 0.63× nominal. The real-data ordering is therefore the same structure-and-size dependence as the simulated one, evaluated at the real cohorts' operating points rather than an unexplained discrepancy; the in-sample mechanism (own-basis ≫ other-basis) holds at every point, and it reproduces on real within-population data at the PC-dominant magnitude (Section 9). The decrease is strict at n=800 and 1600 (at n=300 both attacks sit near ceiling and the gap is within the sweep's tolerance of flat). A Dirichlet admixture sweep (varying the concentration parameter, which interpolates between nearly-pure individuals and near-uniform mixes; it does *not* isolate discrete-vs-continuous structure) again moves the gap with the realized structure, from −0.12 at the most concentrated setting to +0.10 at the most uniform (`s22_admixture.py`), i.e. the ordering tracks structure strength rather than the discrete/admixed label.

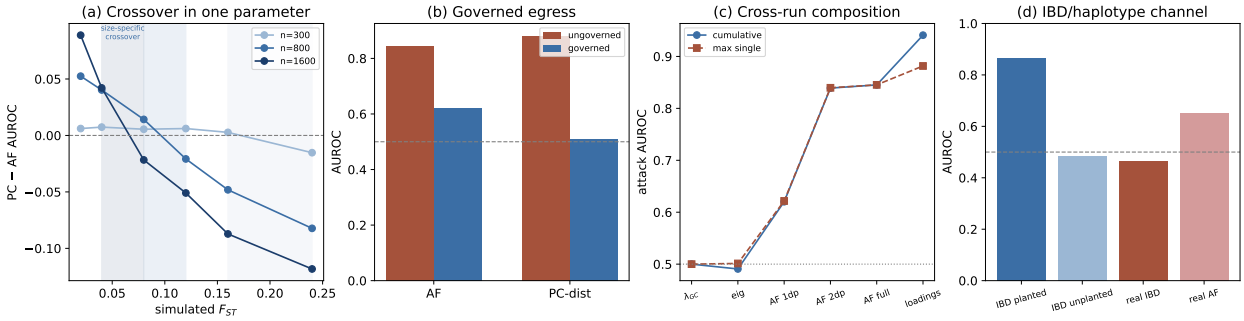


Figure 3: Governance layer and model-dependence. (a) The PC−AF gap decreases in simulated F_{ST} (three site sizes) and crosses zero at a measured structure that depends on site size (measured crossover ≈ 0.03–0.15 across the three sizes; the bands mark the per-size nominal crossing interval). (b) Governed egress: the monitor's manifest-permitted release (AF@1dp, loadings withheld) drops the leakage attacks to the governed tier. (c) Cross-run composition: individually coarsened releases compose above the largest single release. (d) An IBS window-sharing channel detects planted IBD but is at chance unplanted and on real data.

The sanitized-diagnostics paradox, as a result. Replacing the earlier probe-count statistic with one measured on the *released object* (the attack AUROC of the finest AF the analyst actually sees), and running seven analyst families over 12 seeds with a quality-parity margin, the paradox holds in the predicted regime and only there: pre-providing a coarse sanitized block reduces exposure by $\Delta_E = +0.24$ (95% CI $[0.23, 0.26]$) when the sanitized precision is coarser than the analyst's stopping precision, and $\Delta_E = 0$ when it is not—while a *null* analyst that needs no diagnostic shows *no* paradox (the falsification control). The effect is tightly the exposure-curve difference $\Delta_E = e(p_{\text{stop}}) - e(p_{\text{san}})$: the families set only the stopping rung, and quality is nearly flat in the chosen k (so parity is a consequence, not a coincidence); moreover the null family at $p_{\text{san}} \geq 2$ shows a *negative* $\Delta_E \approx -0.24$ because a coarser-than-needed sanitized block *is* the leak—the sharp version of the "precision-gap effect" the prediction names (`s25_paradox.py`).

An IBS window-sharing channel. The haplotype/LD channel above (0.99) is an in-sample-fitted joint table—it transfers at chance. A genuinely distinct identity-by-state (IBS) *window-sharing* attack (the fraction of 8-marker windows a target matches to any study vs any reference individual; run here on unphased dosages—the 1000G VCFs are phased, and `real_phased.py` loads the haplotypes, but this arm has not yet exploited H0/H1 separately) detects a *planted* IBD segment at 0.866 but is at chance on unplanted simulation (0.486) and on real chr22 (0.463, below the real AF channel 0.653; Fig. 3(d)). So the 0.99 adjacent-pair number is a fitted-table artifact, and real haplotype sharing among unrelated 1000G samples leaks little under the tested statistics—a characterization of the tested IBS statistic and the leave-self-out LSR diagnostic, not a bound on the haplotype channel as a whole.

9 Scale, Density, and Ancestry: Where the Ordering Lives

The ordering claims above are operating-point properties; a GPU-ported harness (host-resident int8 panels, blocked exact-Gram PCA on GPU; the governance layer itself is CPU/plaintext) lets us locate the boundary at biobank dimensions and retire three data caveats at once.

The boundary is small ($n \times M$). Sweeping study size and marker count (Fig. 4(a,b)): at n_{sp}=500 (1500 study genomes) the PC channel leads only below $\sim 10^4$ markers (+0.008 at M=5k)

and falls behind above it (-0.02 at $M{=}10^4$, -0.03 at $M{=}5{\cdot}10^4$); by $n_{\mathrm{sp}}{=}2000$ AF leads at every M (-0.03 to -0.18, widening with M); at $n_{\mathrm{sp}}{=}8000$ AF leads decisively (0.75–0.84 vs 0.59–0.65). The dominance is thus a small-study phenomenon, and the AF release saturates first. The mechanism is unaffected by scale: projecting onto the *reference*-fitted basis collapses the PC channel to chance (0.485–0.504) in *every* cell, and the in-sample gap shrinks monotonically with study size within each Fst column (at Fst=0.04: $+0.51 \to +0.15$; Fst=0.08: $+0.50 \to +0.11$, all at matched M). A denser, small-n sweep (Fig. 4(b)) locates the crossing directly.

The gaps are real, and tightly so. A stratified label-permutation null ($2{\cdot}10^4$ permutations/cell, permuting within ancestry) puts both channels far above chance in every cell ($p < 5 \cdot 10^{-5}$), so neither discrimination is a structure artifact; a paired sweep (three 200-seed shards + one 200-seed run for the slow cells, 600 seeds on each fast cell) gives the small-n PC-AF gap as $+0.094$ $[+0.092, +0.096]$, $P(\Delta > 0){=}1.0$ (600 seeds), decaying to $+0.009$ $[+0.008, +0.009]$, $P(\Delta > 0){=}0.94$ at $M{=}8000$ and turning *negative* (-0.014, $P(\Delta > 0){=}0.21$, 200 seeds) at $n{=}1600$. The exact k-ladder (Fig. 4(c)) is monotone with strictly decaying increments ($+.13, +.07, +.06, +.04, +.04, +.03, +.02, +.02, +.01$ for $k{=}1 \ldots 10$ on the simulator)—still climbing slightly at $k{=}10$, so we report it as saturating only in the sense that the per-k increment has fallen to $\lesssim 0.02$; an eigenvalue-only control stays near chance on the simulator, on real chr22/chr1/chr2 (s48/s50 0.38–0.60; s49 0.42–0.50), on msprime (0.50–0.53), and in every cell of the biobank-scale grid (0.497–0.511)—the leak is in the loadings at every tested corpus and scale.

Density does not reverse it. Raising real marker density by 20× (chr22: 2.2k→48k; chr1/2: 13k→264k markers; Fig. 4(d)) leaves the PC channel flat (0.66–0.67) while AF *rises*, so the PC-AF gap widens rather than closes.

Where the ordering lives on real genomes. Running the identical attack code on real 1000G cohorts of varying composition (`s48_real_structure.py`, chr22, 5 split seeds per cohort), the study-fitted PC channel *dominates* AF within a single population and *reaches parity or falls below* it once population composition broadens *and* study size grows: twelve single-population cohorts all give PC > AF (gap $+0.051$ to $+0.136$; GBR 0.915/0.999, CHB 0.884/0.992, TSI 0.862/0.998, CDX 0.885/1.000, …, at $n_{\mathrm{study}}{=}28$–36), two-population mixes give $+0.114$ to $+0.198$, and the 3-/5-super-population sets give $+0.007$ / -0.041 (at $n_{\mathrm{study}}{=}450$ / 715). Each channel's sign is fixed on a random held-out half and the AUROC is evaluated on the *other*, disjoint half (never using the scored individuals' own labels), so control channels sit near chance rather than being folded upward. A stratified label-permutation null (`s54_real_permnull.py`, 5,000 permutations on the evaluated half) places both channels far above chance in every tested cohort ($p \le 2 \cdot 10^{-2}$) and the bootstrap 95% CI over the five per-split gaps excludes zero on the within-population arm (GBR $+0.085$ $[+0.058, +0.126]$, CHB $+0.108$ $[+0.047, +0.202]$, YRI $+0.053$ $[+0.038, +0.070]$; the 3-population cell is near parity and straddles zero, $+0.007$ $[-0.008, +0.020]$, and the 5-population cell is AF-first, -0.041 $[-0.077, -0.005]$; the null tests each channel's separation, not the PC$-$AF difference, and covers the three within-population cohorts plus the 3-/5-population cells rather than the full s48 ladder), and per-cohort split variability is reported in the JSON. The within-population ordering is density-stable: on chr22 the GBR/CHB/YRI cohorts keep PC $\ge$ AF at a ∼17k- and a ∼47k-marker tier (`s55_real_density.py`; gaps $+0.007/+0.035/+0.000$ and $+0.002/+0.021/+0.000$; at these tiny n PC is saturated near 1.0, so the tier comparison is carried by the AF side), so the real within-population M axis is not a single-thinning artifact. The single-population fst_like is 0 by construction (one group), so "within-population" is a definitional operating point, not

a measured structure value; the measured structure rises across the ladder from 0.004 to 0.087. The data points are not monotone in structure alone: the gap is positive through Fst ≤ 0.02, essentially at parity at the 3-population rung (Fst=0.087, gap +0.007), and negative at the 5-population rung (Fst=0.075, gap −0.041), with two local non-monotonic steps, so we bound the crossover only loosely in $[0.02, 0.09]$. The ladder *confounds* structure with study size; a scale-matched rerun (`s48_scalematched.py`, per-group n=20 for every rung, total n_{study}=20/60/100) shows the confound explicitly: at *matched* per-group n, the PC channel still dominates even *across* populations (3-pop, Fst=0.10, gap +0.193, per-split sd 0.020; 5-pop, Fst=0.09, gap +0.137, sd 0.076), while the real cohorts show near parity at n_{study}=450 and AF-first at n_{study}=715; total n thus also grows across the matched ladder, so the data are consistent with—but do not isolate—a joint dependence on total study size and composition. The joint dependence is the same shape the simulator's boundary surface shows, where the crossing value of structure rises as n shrinks (the size-resolved crossing brackets are $[0.027, 0.052]$ at n=1600, $[0.052, 0.077]$ at n=800, $[0.103, 0.151]$ at n=300, `s21_sensitivity.py`; that sweep orients the PC channel by $\max(\text{AUC}, 1 - \text{AUC})$ on the evaluated observations (a direction-agnostic orientation), so its *nulls are biased upward* and its crossing brackets are optimistic upper bounds on the AF-first region; the held-out orientation used in the paired analysis and in the real-data sections removes this bias); the real cohorts sample that surface at small n.

Fixed super-population proportions, growing size. To vary the size axis without changing the composition, we hold the cohort's super-population proportions fixed (equal thirds of the same three super-populations) and sweep the total size (`s51_real_lowstruct.py`, chr22, chr1 and chr2, 5 split seeds per rung). On chr22 the measured structure is constant across the rungs (Fst $\approx$ 0.087–0.091) while the split-mean gap falls monotonically and crosses zero: per-group n=40/80/120/160 gives gap $+0.120/+0.042/+0.007/-0.026$ at total study size 120/240/360/480; chr1 shows the same shape ($+0.058/+0.002/-0.010/-0.018$ at Fst $\approx$ 0.10) and chr2 reaches near-parity at the top rung ($+0.109/+0.059/+0.018/+0.006$ at Fst $\approx$ 0.086). So at *fixed* superpopulation proportions, jointly increasing the study, reference and evaluation panels produces a decreasing split-mean gap on real genomes; the crossing locations on chr22/chr1 ($n_{\text{study}} \approx$ 260–400) are interpolations of the five-split means (the top rungs sit within split noise of zero; chr2 stays positive), and the reference panel grows with the study panel, so this is a joint size effect rather than an isolated study-size effect. A per-seed interpolation of the chr22 crossing is itself unstable: of five split seeds three cross once (mean total study size $\approx$ 341, sd $\approx$ 44), seed 0 crosses three times, and seed 4 does not cross over the tested range (`s60_share_crossing.py`), so the crossing bracket is a five-split-mean statement (the same script finds the top-10 study-fitted loadings capture only $\approx$ 0.28–0.29 of the cohort's total squared deviation). The arm pools each super-population into one group so measured between-group structure is zero by construction; the study cap (n_{study}=167–220 under equal thirds) is a property of that split design, not an absolute 1000G ceiling. The pooled-super-population arm still has PC > AF (chr22 EUR/EAS/AFR $+0.075/+0.051/+0.044$; chr1 +0.110/+0.165/+0.048; chr2 +0.181/+0.111/+0.154; both controls at chance). Growing the study within a single pooled super-population while holding the reference panel at a fixed n_{ref}=150 (so a shrinking reference cannot by itself lower the AF baseline, and the non-members absorb the remainder; `s56_real_poolmax.py`, chr22+chr1+chr2, 5 splits) keeps the gap positive at every study size the pool allows (up to n_{study}=350): the largest cell with a full-size reference is AFR at n_{study}=350, gap +0.102/+0.140/+0.181 on chr22/chr1/chr2 (per-cell sd 0.018–0.037; EUR/EAS reach the same n_{study} with a reference capped at n_{ref}=147–148 and only 6 non-members, so their gaps (+0.062/+0.154 chr22, +0.389/+0.034 chr1, +0.079/+0.065 chr2) are positive but high-variance, sd up to 0.41). The reference-fitted-basis and eigenvalue-only controls stay near chance (0.37–0.63) throughout. So within the largest real structure-zero cohorts 1000G supports the pooled PC>AF

ordering does not flip; a zero-structure real AF-first cohort remains unobserved inside 1000G, and the largest study size at which it is tested is bound by the single-super-population pool. Two independent chromosomes agree on the within/across split (`s50_real_structure_multichrom.py`: chr1/chr2 within-population gaps +0.062 to +0.091, all positive; the three-/five-super-population cells give −0.057 to +0.012, while the EUR two-population mix stays positive (+0.149/+0.150), with the chr2 3-population cell a mild PC > AF exception (+0.012) at the crossing, where the chromosome-level sign is unstable—chr1 is −0.030 there). The same in-sample mechanism operates at this PC-dominant magnitude on real data (`s49_real_mechanism.py`, chr22 and chr1; n_{study}=30–34, k=10, so $k/n \approx 1/3$—the near-1.0 AUROCs are the geometric small-n limit of the in-sample effect, and on the larger simulated slabs the same mechanism gives 0.88 at n=900): the study-fitted loadings recover the leak (top-3 0.91–0.93, full 0.996–0.999), while the reference-fitted-basis control (a disjoint panel's basis, out-of-sample for both scored panels) sits at 0.42–0.52, the eigenvalue-only control (a random orthonormal basis) at 0.42–0.50, and a held-out-half basis—fit on one study half, scored on the *other* half, with no overlap—transfers at chance (0.44–0.60) against a same-half basis of 1.000. So the PC-dominance ordering is not a simulator artifact: it is reproduced on real genomes at the operating points the small-($n\times M$) boundary predicts, with the mixed three-super-population case sitting near the crossing for its size (gap +0.007).

A released haplotype set leaks membership trivially; leave-self-out LSR is a relatedness diagnostic. A genuinely phased attack (Bu et al.) had been left to future work. *First*, a direct check: the study releases its haplotype set, so a member's own two haplotypes are literally present in it. Scoring each candidate by whether its own two haplotypes appear verbatim in the release separates members from non-members perfectly (`s41_directmatch.py`: AUROC 1.00 on chr1/chr2/22, all members present, no non-member present). Releasing a raw haplotype set is therefore trivially membership-leaking, and we do not treat it as a governed object. *Second*, separating the self-match from any signal, we ask whether a member can be detected through a long *shared* segment to *other* released individuals: computing the pairwise longest shared run between each candidate's haplotypes and the released set *with the candidate's own two haplotypes removed* (a leave-self-out relatedness diagnostic), a planted-IBD control is detected (0.671) but the real unrelated-member statistic is at chance (0.486), *below* the AF release—so, leaving aside the self-match, this LSR statistic does not separate unrelated 1000G members. Extending this from chr22 at one marker density to *three* chromosomes (chr1, chr2, chr22), uniformly strided across the surveyed regions (chromosome 1 to 83 Mb, chromosome 2 to 72 Mb, chromosome 22 to 50 Mb), with the shared-run measured in *centiMorgans* via the 1000G IMPUTE2 recombination map rather than in markers (`s41_phased_ibd_genome.py`, 50k markers per chromosome, uniformly strided from the full-density parse over chromosome 1 to 83 Mb / chromosome 2 to 72 Mb / chromosome 22, 3 splits), the planted 10 cM control is detected on every chromosome (0.769/0.766/0.749) and the leave-self-out real arm stays near chance and *below* the AF release (0.512/0.520/0.471 versus AF 0.866/0.864/0.838; the AF-first gap is +0.34 to +0.37)—this leave-self-out longest-shared-run statistic fails to separate unrelated 1000G members from non-members at genetic (centiMorgan) length. This characterizes one specific leave-self-out relatedness statistic; it is *not* a bound on the membership leak of a released haplotype set, which the direct check above shows is total. Two further leave-self-out haplotype statistics (`s52_haplo_second.py`: the max whole-haplotype IBS share of a candidate, and the count of released haplotypes whose IBS share exceeds 0.78, on chr22, chr1 and chr2) are also at chance on the real unrelated arm (0.488/0.496 on chr22, 0.497/0.497 on chr1, 0.497/0.498 on chr2) and detect a planted 1500-marker copied segment (0.798/0.797 on chr22, 0.776/0.788 on chr1, 0.772/0.779 on chr2), so the negative does not depend on the longest-shared-run estimator.

Missingness distinguishes members by batch association, not by the release. The membership-correlated missingness channel was earlier a hand-planted per-marker rate excess. Rebuilding it on the real 1000G dosage skeleton, with processing batches keyed on the real 1000G population label and the member-correlation an explicit batch-composition parameter ρ (`s42_missingness_realistic.p`), the call-rate-table attack rises in ρ: AUROC 0.490, 0.629, 0.755, 0.893, 1.000 at ρ=0, 0.25, 0.5, 0.75, 1 (5 seeds; per-point sd 0.017–0.022 for the interior points, 0 at the saturated endpoint). The call patterns are drawn from a generative model (the real skeleton supplies only the marker set and population labels, not measured missingness), so this reads as a sensitivity curve over the batch parameter rather than a measurement of realized batch effects. Crucially, a candidate can already separate itself by its own total call count, *without any released table or reference table*: that candidate-only baseline scores 0.529/0.651/0.760/0.888/1.000 at the same ρ values, tracking the released-table attack to within 0.021 at every interior point (widest at ρ=0.25: 0.651 vs 0.629), marginally *below* it at ρ=0.75 (0.893 vs 0.888), and both at chance at ρ=0. The missingness result is therefore *batch-associated discrimination* (the member-heavy batch is a property of the individual, observable before any release), not incremental exposure caused by releasing the call-rate table; the batch parameter ρ is the common lever on both curves. The ρ=0 null sits at chance (0.490 for the table attack), and a per-individual call-rate-bonus sweep exposes the magnitude scale (0.49→0.57→0.84→1.00 over bonus 0, 0.005, 0.01, 0.02); within the tested range the base missingness level leaves both curves saturated at ρ=1. A standalone missingness AUROC of 1.00 here does not establish incremental *release* leakage: the candidate-only Baseline matches it, so this construction does not isolate a conditional information gain from the table. (In the multi-object bundle, where a candidate's call count is already available, the released table does add a small conditional increment—+0.008—reported in Section 8.) The channel is therefore reported as a confounding-control result rather than a governance-expanding release family.

A biobank cohort with real haplotype structure. The real-data arm is capped at the 1000G sample set, so the real PC<AF attenuation could in principle be a small-n artefact. We build cohorts far larger than 1000G that carry *real* haplotype structure, by Li–Stephens mosaic haplotype copying from the real 1000G phased library (per-marker switch probability $1 - e^{-\rho\,\mathrm{gap_{bp}}}$; membership labels assigned after generation by the same three-way split, so there is no label circularity) (`semi_synthetic.py`, `s37_realsynth_scale.py`). The study-fitted PC basis is estimated on a train subset of at most 20,000 study rows per cell (the GPU Gram is $n_{\mathrm{fit}} \times n_{\mathrm{fit}}$); at $n_{\mathrm{sp}} \geq 8000$ this is a strict subset of the $3n_{\mathrm{sp}}$ study rows, and we report the fitted fraction per cell, so those cells measure a *subsampled*-fit PC; a fit-size sweep (`s37_fitsens.py`) at two large cells leaves the AF-first gap essentially unchanged across fit fractions: at n_{sp}=8000 the gap is −0.034 (sd 0.008) at every n_{fit} in {5000, 10000, 20000, 24000} (full fit, n_{fit}=n_{site}), and at n_{sp}=16000 it is −0.026 at $n_{\mathrm{fit}} \in \{5000, 10000, 20000\}$ (the full 48000-row fit was not run). Applying the same attack battery (AF-only, train-half-calibrated PC, a joint AF+PC logistic `combined` column, reference-fitted-basis and eigenvalue-only controls; the joint logistic `combined` column is reported as an AUROC and adds the PC feature on top of AF only when `combined` − `af_only` > 0, which it does not here; 5 seeds per cell), AF leads the PC channel at *every* cell from n_{sp}=150 to n_{sp}=16000 (the PC−AF gap runs −0.03 to −0.09; per-cell sd up to about 0.03), and the in-sample gap collapses toward zero as n grows (0.20→0.08→0.03→0.00). The cohort controls hold where they are measurable: the AF spectrum and the top-10 eigenvalue fraction match the donor library at every cell, and LD decay $r^2(d)$ matches at the five cells whose marker panel supports its shortest bin (the two shortest panels, M=3000/6000, have no markers in that bin, so the statistic is undefined there, not mismatched); pairwise relatedness sits within ∼10% of the donor library at six of seven cells (the

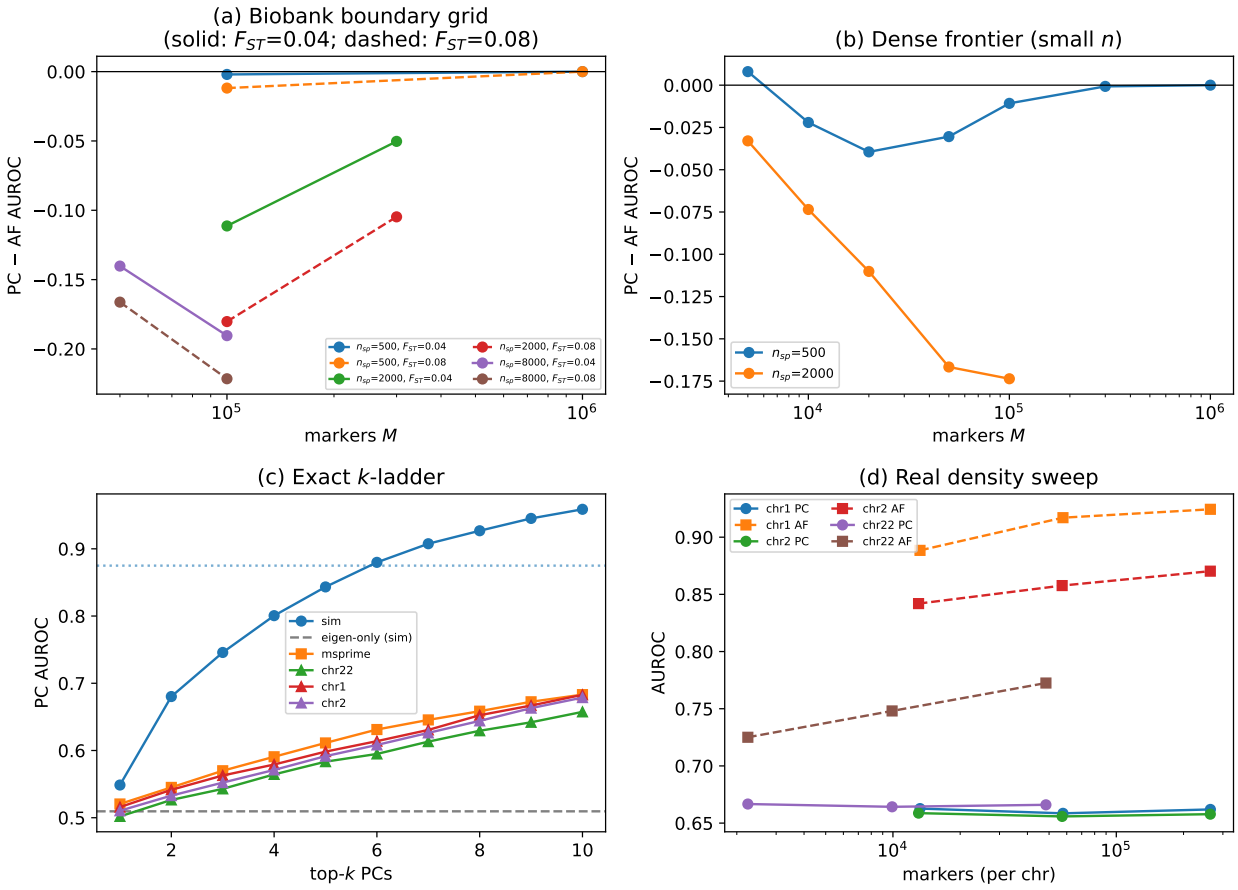


Figure 4: Where the ordering lives. (a) Biobank-scale boundary grid. The PC−AF AUROC gap for $n_{\text{sp}} \in \{500, 2000, 8000\}$; Fst is distinguished by line style (solid F_{ST}=0.04, dashed F_{ST}=0.08), so each curve is a single condition. The gap is negative for $n_{\text{sp}} \geq 2000$. (b) Dense small-n frontier: PC leads only at tiny M. (c) Exact k-ladder on sim/real/msprime: monotone with decaying increments, an eigenvalue-only control at chance. (d) Real marker-density sweep: the PC channel is flat while AF rises, so the gap widens with density.

smallest cell, n_{sp}=150, is +11%). That relatedness number is a dosage parsimony-phase proxy; on the *actual* copied mosaic haplotypes the cohort's longest-shared-run relatedness is 2.2–2.5× the donor library's (`s37_phase_control.py`), as expected for a mosaic cohort built by copying donor segments—so the cohort is more related along actual phase than the donor library, and we state the matched-relatedness claim only on the parsimony-phase proxy statistic that the underlying control defines. The AF/PC ordering holds in the tested mosaics; the effect of the actual-phase relatedness excess on the ordering is unresolved (we measure relatedness in two probed cells, we do not vary it). The gap narrows roughly threefold before plateauing, and the largest delivered cell is n_{sp}=16000 at M=3000 markers (48,000 study genomes, ≈19× the 1000G arm; n_{sp}=16000 is real haplotype structure at the largest delivered cell under this implementation); with real haplotype structure at this scale the AF-first ordering is reproduced, so the real PC<AF attenuation tracks the structured, larger-($n \times M$) regime rather than a hard sample-size cap.

10 Discussion: What Governance Must Cover

The results reframe the governance target. Pipelines secure the final summary statistic while treating diagnostics as free; we show the PC-loading diagnostic is the larger *intermediate* leak relative to the unconditioned AF-LRT baseline in the small-($n \times M$) regime, which holds on real genomes too (within a population the PC channel dominates AF; on real data the AF-first switch tracks total study size as well as composition), and that per-site frequency release is the largest marginal-frequency single channel. An in-sample-fitted haplotype/LD table ranks high in its own construction (but a raw haplotype set is trivially leaking and out of the governed set), and the batch-associated missingness curve is a confounding control rather than a release family. A pre-registered reveal manifest must therefore treat the PC spectrum as a governed object, not a QC byproduct—and must govern the *loadings*, since within our tested attack an eigenvalue-only (random-basis) projection, and fixed data-independent bases including an eigenvalue-weighted one, are at chance (0.46–0.53) while the full study-fitted loadings recover the leak. Cross-run, individually manifest-permitted releases *compose*: a sequence of coarsened single-object releases (a λ_{GC} value, eigenvalues, AF at increasing precision, then the loadings) raises the cumulative attack AUROC to 0.941 versus 0.882 for the largest single release (5/5 seeds; Fig. 3(c)), so a manifest needs an explicit anti-composition clause, which our ledger records (from producer-asserted metadata) as a refinement relation (Section 8). Measuring a constructed multi-object bundle instead of a declared sequence, a joint attacker exceeds the strongest single object by +0.011 in 5/5 seeds, with the excess attributable to orthogonal per-individual channels while the per-panel-constant objects carry no feature for the tested attacker (Section 8), so the anti-composition clause must cover composition across channels, not only a single refinement relation.

Limitations. Absolute numbers are model-specific and the mechanism's magnitude is dimensionality-dependent; the PC-dominance ordering is a small-($n \times M$) regime property, observed on real 1000G within a population (gap +0.05 to +0.14) and switching to AF-first once study size and composition both grow (Section 9); it also fails at the larger simulated site sizes of Fig. 2(d), where AF saturates first (Fig. 3(a) shows the ordering is approximately decreasing in measured structure under a size-controlled F_{ST} sweep; the real cohorts add a study-size confound which the scale-matched rerun exposes). A GPU-ported biobank-scale boundary grid (Section 9) locates the crossover quantitatively. The biobank-scale cohort arm (a semi-synthetic cohort built from real 1000G haplotypes, Section 9) reproduces the AF-first ordering at n_{sp} up to 16000 (48,000 study genomes) with real LD and AF controls that match the donor library and a relatedness control matched on the dosage parsimony-phase proxy (on actual copied phase the cohort is more related than the donor library); the cohort is semi-synthetic, so its absolute magnitudes remain model-specific even though its haplotype structure and proxy relatedness match the donor library. The quantitative claims rest on marginal-frequency attacks (likelihood-ratio, correlation, and learned squared-deviation—forms of the same channel) plus a haplotype/LD channel and an IBS window-sharing proxy channel. The PC channel requires the study's PC loadings to be released (as federated PCA shares them); if loadings are withheld the channel does not apply. Our governance prototype is a plaintext, release-object-level implementation (manifest, egress monitor, hash-chained ledger); the *sole-egress binding* is realized as a process-isolation runtime with a PEP-578 audit hook, a single sanctioned pipe channel, and the child's stdout/stderr discarded, tested against the careless-author bypass under a threat model of ordinary Python with no native code and no root—it is not an MPC/SPU implementation, and a local user with process control remains outside the model. GPU acceleration was used only for the leak-measurement kernels; the GPU harness's rank-based AUROC does not apply midrank tie handling, but a representative archived cell has zero exact ties (3000/3000 unique

scores; `s59_auroc_tiecheck.py`) and the CPU path uses midrank everywhere, so the checked values are unaffected. The biobank-scale simulated cells use a Gaussian diffusion approximation of the Balding–Nichols per-population frequency draw (matching BN to $O(\mathrm{Fst}^2)$) and a GPU-side LD construction that differs in detail from the CPU simulator (a shared latent flipping whole blocks vs. CPU per-marker flips); the exact Beta draw is used in all small-scale cells, and the independent msprime and real-1000G arms are used as external checks. The cohort arm is semi-synthetic (mosaic copying from real 1000G haplotypes); the per-individual metric's per-sample magnitudes are resolution-limited and are reported as an in-sample, resolution-limited trade-off summary rather than an absolute privacy guarantee (ordering-only on the real high-AUROC channels).

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